Cellula NLP — RNN | Updated Dataset

Reorganized report from saved study.ipynb outputs

Dataset: cellula_3000_AUGMENTED_PROVISIONAL.csv; 3,000 rows; 9 classes. Input combines cleaned query and image description. Stratified split: train 2,400 / validation 300 / test 300. Tokenizer fitted on training only. Sequence length 44.

Metric	Saved validation result
Accuracy	95.00%
Macro F1	0.9219
Weighted F1	0.9439

Architecture: Embedding(128) → SimpleRNN(64) → Dropout(0.3) → Dense(9, softmax). Loss: sparse categorical crossentropy; optimizer: Adam (learning rate 0.001); batch size: 32; early stopping on validation loss.

Important evaluation limitation

These figures are preserved from the uploaded study notebook; this package did not retrain the models. They reflect a random validation split of augmented data. Exact-row deduplication alone does not rule out paraphrase/template overlap, and augmentation may contain questionable labels. These results must not be represented as leakage-free test performance.

Reproduction: open the matching notebook, verify the augmented CSV path, Run All, inspect classification report/confusion matrix, and export deployment files from the LSTM notebook.